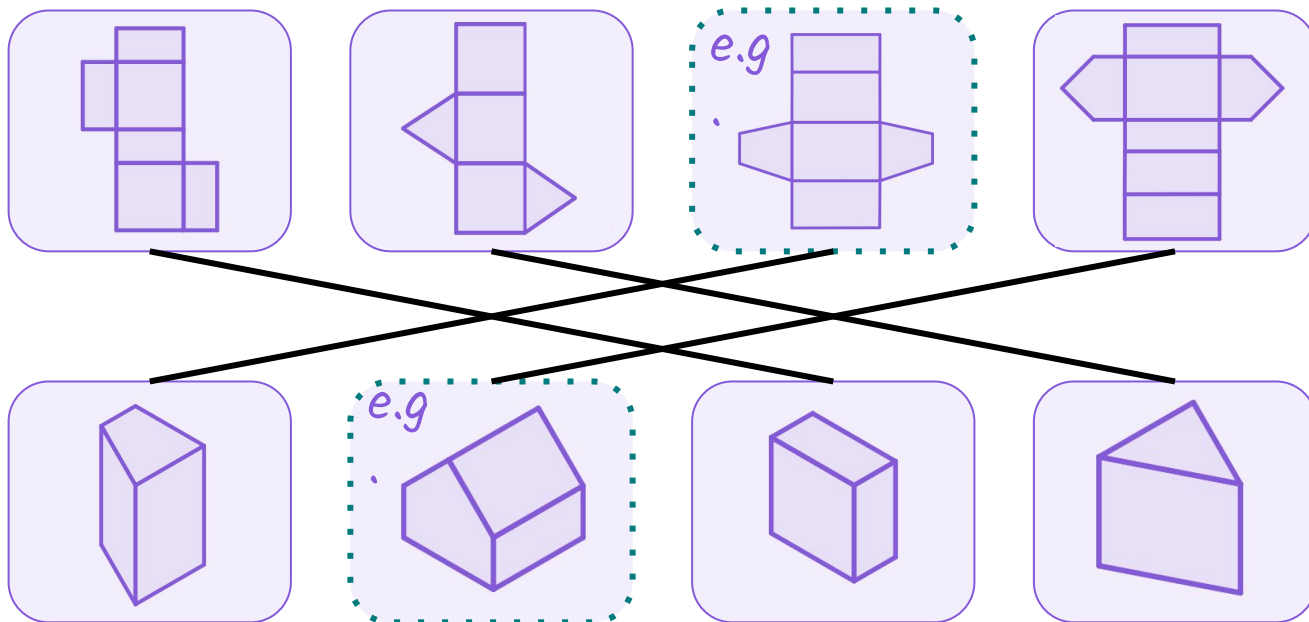




Surface area of prisms

Task A: Nets of right prisms

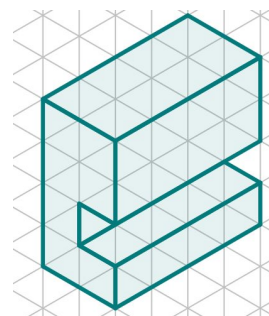
1) Match the net to its prism. Sketch the missing net and prism.



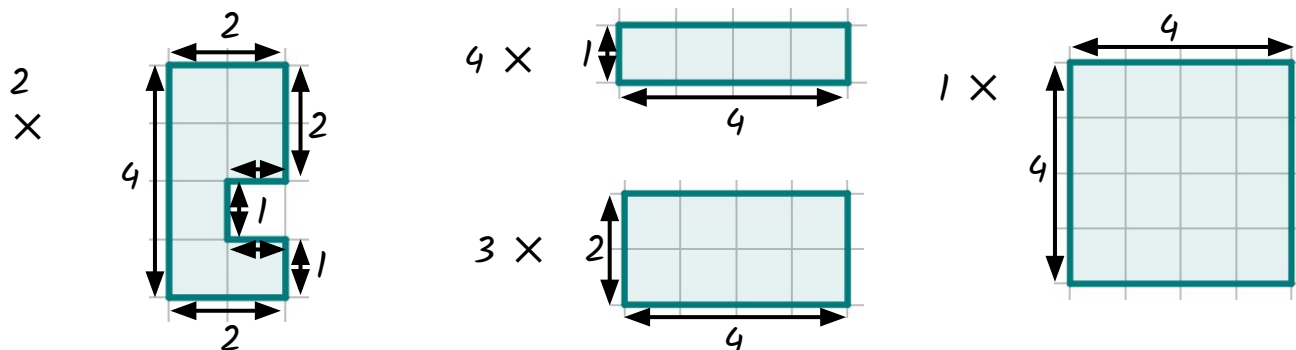
Task B: Calculating the surface area of prisms

1) a) Complete the description of the faces for this right octagonal prism.

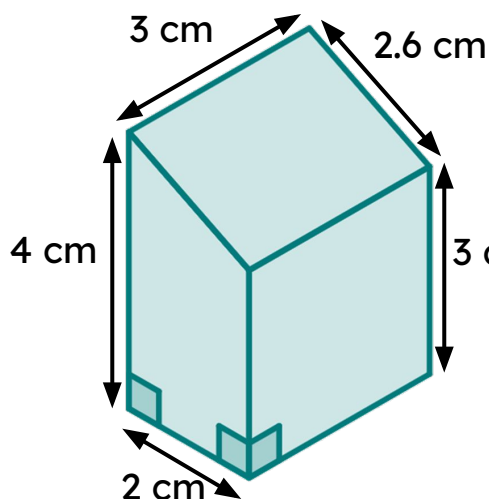
- 2 irregular octagons
- 8 rectangles



b) Draw all the **unique** faces of this prism, with dimensions.



2) Work out the surface area of this right trapezium based prism.



$2 \times \text{area of each trapezium}$

$$= 2 \times \left(\frac{1}{2} \times (4 + 3) \times 2 \right) = 14 \text{ cm}^2$$

$$\text{rectangle 1} = 2 \times 3 = 6 \text{ cm}^2$$

$$\text{rectangle 2} = 3 \times 3 = 9 \text{ cm}^2$$

$$\text{rectangle 3} = 3 \times 2.6 = 7.8 \text{ cm}^2$$

$$\text{rectangle 4} = 3 \times 4 = 12 \text{ cm}^2$$

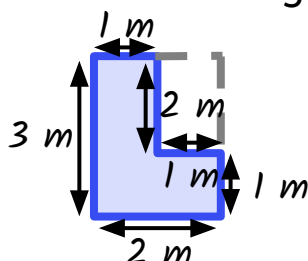
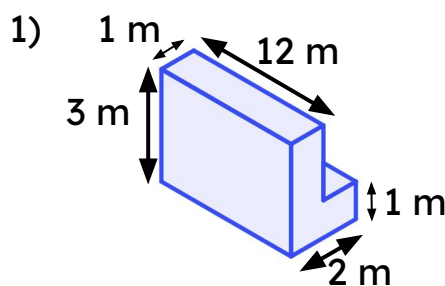
Surface area

$$= 14 + 6 + 9 + 7.8 + 12 = 48.8 \text{ cm}^2$$

Task C: Being efficient in finding the surface area

Calculate the surface area of the following right prisms

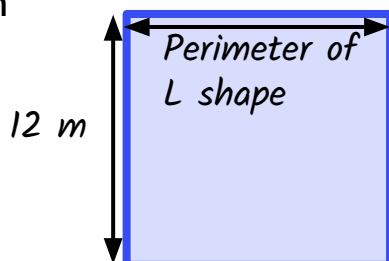
Diagrams not drawn to scale



$2 \times \text{Area of L shape}$

$$= 2 \times ((3 \times 2) - (2 \times 1))$$

$$= 2 \times 4 = 8 \text{ m}^2$$

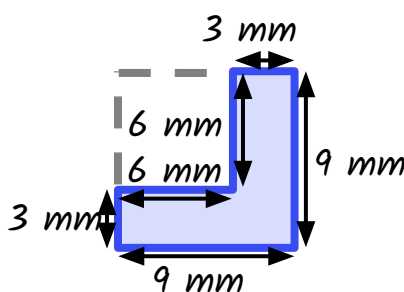
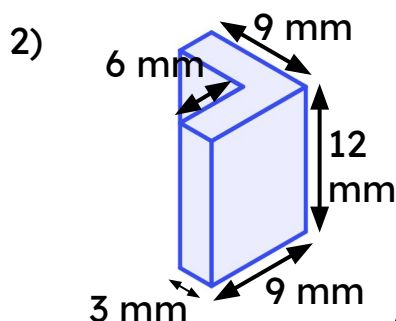


$12 \times \text{perimeter of L shape}$

$$= 12 \times (3 + 1 + 2 + 1 + 1 + 2)$$

$$= 12 \times 10 = 120 \text{ m}^2$$

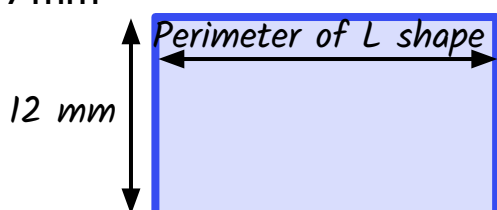
$$\text{Surface area} = 128 \text{ m}^2$$



$2 \times \text{Area of L shape}$

$$= 2 \times ((9 \times 9) - (6 \times 6))$$

$$= 2 \times 45 = 90 \text{ mm}^2$$



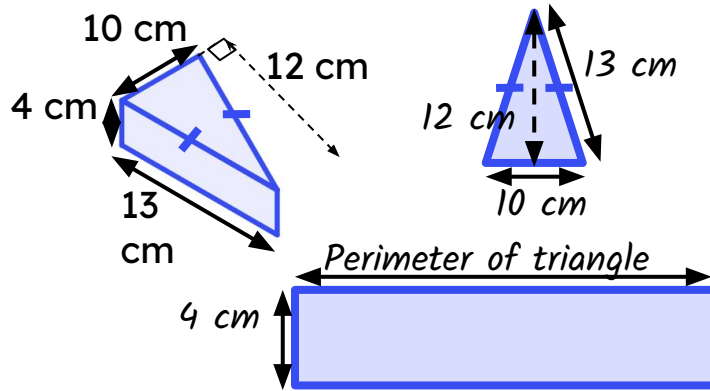
$12 \times \text{perimeter of L shape}$

$$= 12 \times (9 + 3 + 6 + 6 + 3 + 9)$$

$$= 12 \times 36 = 432 \text{ mm}^2$$

$$\text{Surface area} = 522 \text{ mm}^2$$

3)

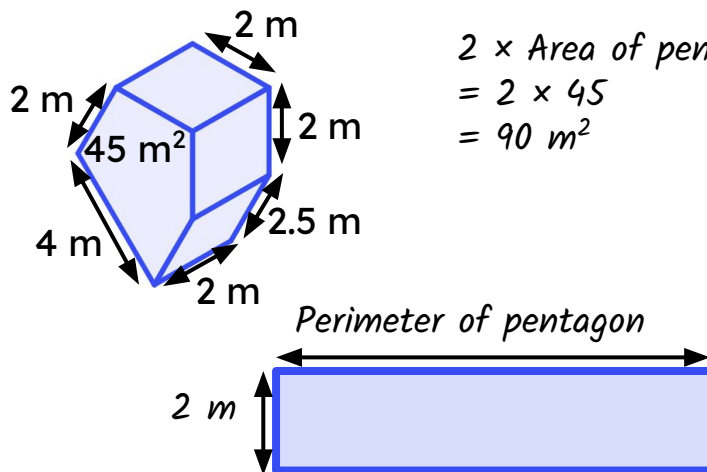


$$\begin{aligned} &2 \times \text{Area of triangle} \\ &= 2 \times \left(\frac{1}{2} \times 10 \times 12 \right) \\ &= 120 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} &4 \times \text{perimeter of triangle} \\ &= 4 \times (13 + 10 + 13) \\ &= 4 \times 36 = 144 \text{ cm}^2 \end{aligned}$$

$$\text{Surface area} = 264 \text{ cm}^2$$

4)



$$\begin{aligned} &2 \times \text{Area of pentagon} \\ &= 2 \times 45 \\ &= 90 \text{ m}^2 \end{aligned}$$

$$\begin{aligned} &2 \times \text{perimeter of pentagon} \\ &= 2 \times (2 + 4 + 2.5 + 2 + 2) \\ &= 2 \times 12.5 = 25 \text{ m}^2 \end{aligned}$$

$$\text{Surface area} = 115 \text{ m}^2$$

